# Lorentz transformation of EM field

Found $F_{\alpha\beta} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & B_x & 0 \end{pmatrix}$

How does $F$, and $\vec{E}$ and $\vec{B}$, transform?

$$F^{\alpha\beta} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}$$

The rule $F'^{\mu\nu} = L^{\mu}{}_{\alpha} L^{\nu}{}_{\beta} F^{\alpha\beta}$

while $F'_{\mu\nu} = (L^{-1})^{\alpha}{}_{\mu} (L^{-1})^{\beta}{}_{\nu} F_{\alpha\beta}$ (see Lect 3, Aug 27)

$$L^{\mu}{}_{\alpha} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \text{while} \quad L^{-1} = \begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

For example: $F'_{01} = (L^{-1})^{\alpha}{}_{0} (L^{-1})^{\beta}{}_{1} F_{\alpha\beta}$

Note that $(L^{-1})^{\alpha}{}_{0}$ $\alpha \leftarrow$ row, $0 \leftarrow$ column

Sum over $\alpha$: $F'_{01} = (L^{-1})^{0}{}_{0} F_{0\beta} (L^{-1})^{\beta}{}_{1} + (L^{-1})^{1}{}_{0} F_{1\beta} (L^{-1})^{\beta}{}_{1}$

Other elements of $(L^{-1})^{\alpha=2,3}{}_{0} = 0$. Now sum over $\beta$:

$$\Rightarrow F'_{01} = (L^{-1})^{0}{}_{0}\Big((L^{-1})^{0}{}_{1}\underbrace{F_{00}}_{=0} + (L^{-1})^{1}{}_{1} F_{01}\Big) \qquad (L^{-1})^{\beta=2,3}{}_{1} = 0$$

$$+ (L^{-1})^{1}{}_{0}\Big((L^{-1})^{0}{}_{1} F_{10} + (L^{-1})^{1}{}_{1}\underbrace{F_{11}}_{=0}\Big)$$

$$= \gamma\left(\gamma\, E_x/c\right) + \gamma\beta\left(\gamma\beta(-E_x/c)\right)$$

$$= (\gamma^2 - \gamma^2\beta^2)\, E_x/c = E_x/c$$

But $F'_{01} = E'_x/c \Rightarrow E'_x = E_x$.

A simpler way is to recognize that $F_{\alpha\beta}$ transforms as $A_\alpha B_\beta$, outer product of two covariant vectors:

$$A'_\alpha = (L^{-1})^{\ell}{}_{\alpha} A_\ell$$

Then $A'_0 = (L^{-1})^{\ell}{}_{0} A_\ell = \gamma A_0 + \gamma\beta A_1$

$$B'_1 = (L^{-1})^{t}{}_{1} B_t = \gamma\beta B_0 + \gamma B_1$$

$$\Rightarrow F'_{01} \sim A'_0 B'_1 = \gamma(A_0 + \beta A_1)\gamma(\beta B_0 + B_1)$$

$$= \gamma^2\left(\beta A_0 B_0 + A_0 B_1 + \beta^2 A_1 B_0 + \beta A_1 B_1\right)$$

$$= \gamma^2\Big(\beta \underbrace{F_{00}}_{=0} + F_{01} + \beta^2 \underbrace{F_{10}}_{=-F_{01}} + \beta \underbrace{F_{11}}_{=0}\Big)$$

$$= \gamma^2(1-\beta^2)\, F_{01} \text{ as before.}$$

$$F'_{02} \sim A'_0 B'_2 = \gamma(A_0 + \beta A_1) B_2 = \gamma(A_0 B_2 + \beta A_1 B_2) =$$

$$= \gamma(F_{02} + \beta F_{12})$$

$$F'_{02} = \frac{E'_y}{c} = \gamma\left(\frac{E_y}{c} + \beta(-B_z)\right)$$

$$\Rightarrow E'_y = \gamma(E_y - v B_z)$$

Similarly we find $E'_z = \gamma(E_z + v B_y)$

$$B'_x = B_x$$

$$B'_y = \gamma\left(B_y + \frac{v}{c^2} E_z\right)$$

$$B'_z = \gamma\left(B_z - \frac{v}{c^2} E_y\right)$$

(check this on your own as part of homework problem)

Thus, field along $\vec{v}$ does not transform $(E_x, B_x)$ while that perpendicular to it – does.

$$\begin{cases} \vec{E}'_\perp = \gamma(\vec{E}_\perp + \vec{v} \times \vec{B}) & \vec{E}'_\parallel = \vec{E}_\parallel \\ \vec{B}'_\perp = \gamma\left(\vec{B}_\perp - \frac{1}{c^2} \vec{v} \times \vec{E}\right) & \vec{B}'_\parallel = \vec{B}_\parallel \end{cases}$$

**Important Note.**

As written, the transformation formula in the question is suppressing an important subtlety. The field strength tensor is a tensor *field*; it is a function on spacetime. As such, it's not just its indices that must transform under a Lorentz transformation, but also its argument. The full transformation is as follows:

$$F'^{\mu\nu}(x') = \Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta}F^{\alpha\beta}(x), \qquad x' = \Lambda x$$

Or, put another way

$$F'^{\mu\nu}(x) = \Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta}F^{\alpha\beta}(\Lambda^{-1}x)$$

Authors often suppress this fact in the notation, but I found it confusing when I first learned the subject, so I think it's dangerous to not be explicit.

As a terminological aside, the transformation of the tensor indices is often referred to as a **target space transformation** while the transformation of its argument is sometimes referred to as a **spacetime transformation**.

**Target space transformation - efficient computation**

An efficient way to compute an arbitrary target space transformation in this context is to note that when dealing with strings of contracted two-tensors, one can convert the expression into a sequence of matrix multiplications.

If we define matrices $\Lambda = (\Lambda^{\mu}{}_{\nu})$ and $F = (F^{\mu\nu})$ where the left-most index is the "row" label and the right-most index is the "column" label, then we find that

$$\Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta}F^{\alpha\beta} = \Lambda^{\mu}{}_{\alpha}F^{\alpha\beta}\Lambda^{\nu}{}_{\beta} = (\Lambda F \Lambda^{T})^{\mu\nu}$$

where the expression in parenthesis denotes successive matrix multiplication.

**Special Cases**

The matrix multiplication above is made significantly easier provided the Lorentz transformation one is performing is special. In particular, suppose for instance that the Lorentz transformation is a boost along the $x$-direction. Then the matrix $\Lambda$ will have the following block-matrix form

$$\Lambda = \begin{pmatrix} \lambda & 0 \\ 0 & I_2 \end{pmatrix}$$

where

$$\lambda = \begin{pmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{pmatrix}, \qquad I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Therefore, if we write $F$ is 2-by-2 block matrix form;

$$F = \begin{pmatrix} f_{11} & f_{12} \\ f_{12} & f_{22} \end{pmatrix}$$

then we have

$$\begin{aligned} F' &= \begin{pmatrix} \lambda & 0 \\ 0 & I_2 \end{pmatrix}\begin{pmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{pmatrix}\begin{pmatrix} \lambda & 0 \\ 0 & I_2 \end{pmatrix} \\ &= \begin{pmatrix} \lambda f_{11} & \lambda f_{12} \\ f_{21} & f_{22} \end{pmatrix}\begin{pmatrix} \lambda & 0 \\ 0 & I_2 \end{pmatrix} \\ &= \begin{pmatrix} \lambda f_{11}\lambda & \lambda f_{12} \\ f_{21}\lambda & f_{22} \end{pmatrix} \end{aligned}$$

so now one just needs do perform four 2-by-2 matrix multiplications, which is relatively easy. Boosts in the $y$- and $z$- directions are similarly straightforward.

$$\Lambda^{\mu}{}_{\alpha} F^{\alpha\beta} \Lambda^{\nu}{}_{\beta}$$

left - row
right - column

$$A_{ij} B_{ik}$$

$$(\Lambda F \Lambda^T)^{\mu\nu}$$

$$\underbrace{\Lambda^{\mu\alpha} F^{\alpha\beta}}_{O^{\mu\beta}} (\Lambda^T)^{\beta\nu}$$

$$O^{\mu\beta} (\Lambda^T)^{\beta\nu}$$

$$(\Lambda^T)^{\beta}{}_{\nu} = \Lambda^{\nu}{}_{\beta}$$

In[783]:=
```
L = {{γ, -γβ, 0, 0}, {-γβ, γ, 0, 0}, {0, 0, 1, 0}, {0, 0, 0, 1}}
MatrixForm[L]
F = {{0, -Ex / c, -Ey / c, -Ez / c},
  {Ex / c, 0, -Bz, By}, {Ey / c, Bz, 0, -Bx}, {Ez / c, -By, Bx, 0}}
MatrixForm[F]
MatrixForm[L.F.L] /. {γ^2 → 1 / (1 - β^2)} // Simplify
% /. {c β → v} // Simplify
MatrixForm[%]
```

Out[783]= $\{\{\gamma, -\beta\gamma, 0, 0\}, \{-\beta\gamma, \gamma, 0, 0\}, \{0, 0, 1, 0\}, \{0, 0, 0, 1\}\}$

Out[784]//MatrixForm=

$$\begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Out[785]= $\left\{\left\{0, -\frac{Ex}{c}, -\frac{Ey}{c}, -\frac{Ez}{c}\right\}, \left\{\frac{Ex}{c}, 0, -Bz, By\right\}, \left\{\frac{Ey}{c}, Bz, 0, -Bx\right\}, \left\{\frac{Ez}{c}, -By, Bx, 0\right\}\right\}$

Out[786]//MatrixForm=

$$\begin{pmatrix} 0 & -\frac{Ex}{c} & -\frac{Ey}{c} & -\frac{Ez}{c} \\ \frac{Ex}{c} & 0 & -Bz & By \\ \frac{Ey}{c} & Bz & 0 & -Bx \\ \frac{Ez}{c} & -By & Bx & 0 \end{pmatrix}$$

Out[787]//MatrixForm=

$$\begin{pmatrix} 0 & -\frac{Ex}{c} & -\frac{Ey\,\gamma}{c} + Bz\,\beta\,\gamma & -\frac{(Ez+By\,c\,\beta)\,\gamma}{c} \\ \frac{Ex}{c} & 0 & -Bz\,\gamma + \frac{Ey\,\beta\,\gamma}{c} & \left(By + \frac{Ez\,\beta}{c}\right)\gamma \\ \frac{Ey\,\gamma}{c} - Bz\,\beta\,\gamma & \left(Bz - \frac{Ey\,\beta}{c}\right)\gamma & 0 & -Bx \\ \frac{(Ez+By\,c\,\beta)\,\gamma}{c} & -\frac{(By\,c+Ez\,\beta)\,\gamma}{c} & Bx & 0 \end{pmatrix}$$

Out[788]= $\left\{\left\{0, -\frac{Ex}{c}, -\frac{Ey\,\gamma}{c} + Bz\,\beta\,\gamma, -\frac{(Ez + By\,v)\,\gamma}{c}\right\}, \left\{\frac{Ex}{c}, 0, -Bz\,\gamma + \frac{Ey\,\beta\,\gamma}{c}, \left(By + \frac{Ez\,\beta}{c}\right)\gamma\right\}, \left\{\frac{Ey\,\gamma}{c} - Bz\,\beta\,\gamma, \left(Bz - \frac{Ey\,\beta}{c}\right)\gamma, 0, -Bx\right\}, \left\{\frac{(Ez + By\,v)\,\gamma}{c}, -\frac{(By\,c + Ez\,\beta)\,\gamma}{c}, Bx, 0\right\}\right\}$

Out[789]//MatrixForm=

$$\begin{pmatrix} 0 & -\frac{Ex}{c} & -\frac{Ey\,\gamma}{c} + Bz\,\beta\,\gamma & -\frac{(Ez+By\,v)\,\gamma}{c} \\ \frac{Ex}{c} & 0 & -Bz\,\gamma + \frac{Ey\,\beta\,\gamma}{c} & \left(By + \frac{Ez\,\beta}{c}\right)\gamma \\ \frac{Ey\,\gamma}{c} - Bz\,\beta\,\gamma & \left(Bz - \frac{Ey\,\beta}{c}\right)\gamma & 0 & -Bx \\ \frac{(Ez+By\,v)\,\gamma}{c} & -\frac{(By\,c+Ez\,\beta)\,\gamma}{c} & Bx & 0 \end{pmatrix}$$

In[790]:=
```
g = {{1, 0, 0, 0}, {0, -1, 0, 0}, {0, 0, -1, 0}, {0, 0, 0, -1}}
MatrixForm[L.g.L] /. {γ^2 → 1 / (1 - β^2)} // Simplify
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Out[790]= {{1, 0, 0, 0}, {0, -1, 0, 0}, {0, 0, -1, 0}, {0, 0, 0, -1}}

Out[791]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

Consider frame K with $\vec{E} = (0, E, 0) \Rightarrow E_y = E$

$\vec{B} = (0, 0, B) \Rightarrow B_z = B$

$\vec{E} \cdot \vec{B} = 0.$

Now frame K' moving with $\vec{V} = (V, 0, 0)$ relative to K.
Note that $\vec{V} \perp \vec{E} \perp \vec{B}$. ~~and~~

$$\vec{E}' = \gamma\left(\vec{E} + \vec{V} \times \vec{B}\right) \Rightarrow E'_y = \gamma(E - VB), \quad E'_{x,z} = 0$$

$$\vec{B}' = \gamma\left(\vec{B} - \frac{1}{c^2}\vec{V} \times \vec{E}\right) \Rightarrow B'_z = \gamma\left(B - \frac{1}{c^2}VE\right), \quad B'_{x,y} = 0.$$

$$\vec{V} \times \vec{B} = \begin{vmatrix} x & y & z \\ V & 0 & 0 \\ 0 & 0 & B \end{vmatrix} = -\hat{y}\, VB$$

$$\vec{V} \times \vec{E} = \begin{vmatrix} x & y & z \\ V & 0 & 0 \\ 0 & E & 0 \end{vmatrix} = \hat{z}\, VE$$

Thus we still have $\vec{E}' \cdot \vec{B}' = 0 = \vec{E} \cdot \vec{B}$.

$\underline{\underline{E < cB}}$ Note that if $V = \frac{E}{B}$ (<u>so that $E < cB$!</u>)

then $E'_y = 0 \Rightarrow$ in K' we only have magnetic field $B'_z = \gamma\left(B - \frac{1}{c^2}\frac{E}{B}E\right) = \gamma\left(B - \frac{E^2}{cB}\right)$.

$\underline{\underline{E > cB}}$ In this situation we can choose

$V = c^2\frac{B}{E}$ so that $B'_z = 0$ while $E'_y = \gamma\left(E - \frac{c^2B^2}{E}\right)$

$\rightarrow V = c^2\frac{B}{E} < c \Rightarrow \frac{cB}{E} < 1$

However, if $\underline{E = cB}$, then required $V = \frac{E}{B} = \frac{cB}{B} = c$

(or $V = c^2 \frac{B}{E} = c^2 \frac{B}{cB} = c$) so that $\gamma = \infty$

$$E'_y = \gamma \cdot (E - vB) \to \frac{1}{\sqrt{1 - \frac{c^2}{c^2}}} (cB - cB) = \frac{0}{0} \quad ?$$

$$B'_z = \gamma \left(B - \frac{1}{c^2} vE\right) \longrightarrow \frac{1}{\sqrt{1 - \frac{c^2}{c^2}}} \left(B - \frac{1}{c^2} c \cdot cB\right) \quad ?$$

so that $E'_y = c\, B'_z$.

Of course such a frame is not possible.

But if $E = cB$ we have in any other frame $(v < c)$

$$E'_y = \gamma (E - vB) = \gamma (cB - vB) = \gamma (c - v) B$$

$$B'_z = \gamma \left(B - \frac{1}{c^2} vE\right) = \gamma \left(B - \frac{1}{c^2} vcB\right) = \gamma \left(1 - \frac{v}{c}\right) B$$

so that $E'_y = c B'_z$.

Hence in that case $\vec{E}' \cdot \vec{B}' = 0$ and moreover

$E = cB$ means $E' = cB'$ in all possible frames?

<u>But why?</u> Consider invariant $F_{\alpha\beta} F^{\alpha\beta} =$

$$= \left(\frac{E_x}{c}\right)\left(-\frac{E_x}{c}\right) + \left(\frac{E_y}{c}\right)\left(-\frac{E_y}{c}\right) + \left(\frac{E_z}{c}\right)\left(-\frac{E_z}{c}\right)$$

$$+ \left(-\frac{E_x}{c}\right)\left(\frac{E_x}{c}\right) + (-B_z)(-B_z) + (B_y)(B_y)$$

$$+ \left(-\frac{E_y}{c}\right)\left(\frac{E_y}{c}\right) + B_z B_z + (-B_x)(-B_x)$$

$$+ \left(-\frac{E_z}{c}\right)\left(\frac{E_z}{c}\right) + (-B_y)(-B_y) + B_x B_x$$

$$\Rightarrow F_{\alpha\beta} F^{\alpha\beta} = 2\left(\vec{B}^2 - \vec{E}^2/c^2\right)$$

Thus for $E = cB$ $\quad F_{\alpha\beta} F^{\alpha\beta} = 0.$

Therefore, being invariant, it means that

$F'_{\alpha\beta} F'^{\alpha\beta} = 0$ as well $\Rightarrow \vec{B}'^2 - \frac{1}{c^2}\vec{E}'^2 = 0$

$\Rightarrow E' = cB'$ in all frames!

There is yet another invariant — pseudoscalar

$\varepsilon_{\alpha\beta\mu\nu} F^{\alpha\beta} F^{\mu\nu} = \frac{4}{c}\vec{E}\cdot\vec{B}$ (homework problem)

Its invariance means that $\vec{E}\cdot\vec{B} = \vec{E}'\cdot\vec{B}'$.

Hence if $\vec{E}\cdot\vec{B} = 0$ then $\vec{E}'\cdot\vec{B}' = 0$.

* Therefore, if EM fields are non-zero and non-perpendicular in one frame, then there is no frame in which one of them is <u>zero</u>!